FLATNESS AND ITS RELATIVES THROUGH THE PUSHOUT EQUIVALENCE OF SCHEMES OVER A FIBRE SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4402-4423 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4425-4430, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The pushout equivalence setup (lemma-equivalence-categories-schemes-over-pushout, same chapter, collected as S3-morph-equivalence-categories-pushout L1424) and the algebra lemma characterizing finite type/flatness/smoothness/etaleness of algebras over fibre-product rings (More on Algebra); cited inputs with the affine-localization reduction written in the retained span.

Source reference. Original lemma 07RV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4277–4293. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08KQ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4430–4430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In the situation of Lemma 07RV. If $V' = G(V, U', \varphi)$ for some triple (V, U', φ), then*

1. *$V' \to Y'$ is locally of finite type if and only if $V \to Y$ and $U' \to X'$ are locally of finite type,*
2. *$V' \to Y'$ is flat if and only if $V \to Y$ and $U' \to X'$ are flat,*
3. *$V' \to Y'$ is flat and locally of finite presentation if and only if $V \to Y$ and $U' \to X'$ are flat and locally of finite presentation,*
4. *$V' \to Y'$ is smooth if and only if $V \to Y$ and $U' \to X'$ are smooth,*
5. *$V' \to Y'$ is étale if and only if $V \to Y$ and $U' \to X'$ are étale, and*
6. *add more here as needed.*

If W' is flat over Y', then the adjunction mapping $G(F(W')) \to W'$ is an isomorphism. Hence F and G define mutually quasi-inverse functors between the category

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of schemes flat over Y' and the category of triples (V, U', φ) with $V \to Y$ and $U' \to X'$ flat.

Proof. Looking over affine pieces the assertions of this lemma are equivalent to the corresponding assertions of More on Algebra, Lemma 08KQ. □